

Factbook for AI Environmental Damage

Posted by u/PixelSteel • • 0 points • 3 comments

Hey all,

Personally, I've been very sick of the whole "every query is one bottle of water" lie and the amount of lies surrounding energy usage as well. I decided to compile a factbook here for anyone to save this post and use against antis when they decide to spread misinformation.

****Water Usage****

Water usage can actually vary depending on what model you use. You may be familiar with this [Washington Post article] (<https://www.washingtonpost.com/technology/2024/09/18/energy-ai-use-electricity-water-data-centers/>) that incorrectly estimated how much water an AI model (in this case GPT-3) uses. They claimed:

>****Once**** requires ****519**** ****milliliters**** of water,

>a little more than ****1 bottle****

Looking at the bottom of the article's methodology, they utilized[this paper] (<https://arxiv.org/pdf/2304.03271>). "Making AI Less "Thirsty": Uncovering and Addressing the Secret Water Footprint of AI Models." (UC Riverside, et al.) The paper highlights different ways data centers utilize water to train an AI model or for AI query usage. The author of the Washington Post article was clearly bias, as in the same paper we're presented with this table:

[Table 1](<https://preview.redd.it/af1rbdbod1ag1.png?width=1601&format=png&auto=webp&s=66e537261f48b2d604ae2f6c93e7ead49cf7fb3f>)

You'll notice the last column "# of Requests for 500ml Water" or how many requests it'll take to consume 16 ounces of water - equivalent to a bottle of water. The lowest number of requests (or queries) is 17.7 from the data centers in Sweden. In Sweden, it'll cost you roughly ****28.24ml**** of water for one request - or 0.9 ounces. However, in the U.S. average row we see 29.6 requests per 16 oz of water. This comes to ****16.9ml**** or *****half an ounce of water*****.

Then, we can look at other sources. Sam Altman, the CEO of OpenAI stated water usage is about **0.3ml** per query with the recent models. Of course he would have heavy bias in this. Gemini's official page (Google) says each query uses **0.26ml**. These estimates should be considered with a large pile of salt, as they're from companies who heavily invest in AI. If we were to look at another academic study different from UC Riverside, from [IE University](https://www.ie.edu/insights/articles/from-cloud-to-cup-how-much-water-does-your-chatgpt-drink/):

>In 2023, Pengfei Li of the Rochester Institute of Technology and coauthors calculated that every 20 to 50 questions to a large language model like ChatGPT [require the equivalent of a 500 ml bottle of water](https://arxiv.org/abs/2304.03271), mostly for cooling the servers that generate responses. This means that each single prompt may “drink” somewhere **between 10 and 25 milliliters**.

This data is on par with what UC Riverside reported. We can conclude that each request can take around 10 to 20ml. I used 20 instead of the reported 25 due to UC Riverside's report of aprx. 16.9ml. On the low end, each query could be **a third of an ounce** and on the high end each query could be **up to 0.68 ounces**.

Carbon Output

TBD

Local Environment Impact

TBD

I'll work on the other sections of this factbook later, but for now this is what we have for water usage. If an anti tries to tell you AI models are "sucking up lakes" or "consuming bottles of water" tell them it's only 10 to 20ml per request using these facts backed by credible sources.

Comments

u/Additional_Boot_8935 • 2 points • 2025-12-29 07:24 UTC

It's such a silly talking point. The combination of silly climate alarmists, anti-AI gatekeepers, and religious zealots who also hate AI is such a toxic brew.

They have no understanding of the technology behind the cooling systems, the recapturing of water use, etc...

But, even for pretend sake, let's say AI was using up fresh water resources at alarming rates. Well, then we would simply advance the timeline on desalination technology (which is already happening at some universities). Water is the most abundant resource on the earth's surface, there is no shortage, and if AI led to sped up advancements in desalination and turning unusable water into potable water - that would be a great thing for the inhabitants of planet earth.

u/Serasugee • 2 points • 2025-12-29 06:09 UTC

Not to mention these are the online services. Many run it locally, but people ignore that.

u/Whilpin • 1 points • 2025-12-29 00:45 UTC

so... I'm ****NOT**** burning down forests that make pencils? 🤔